



MakanPrint

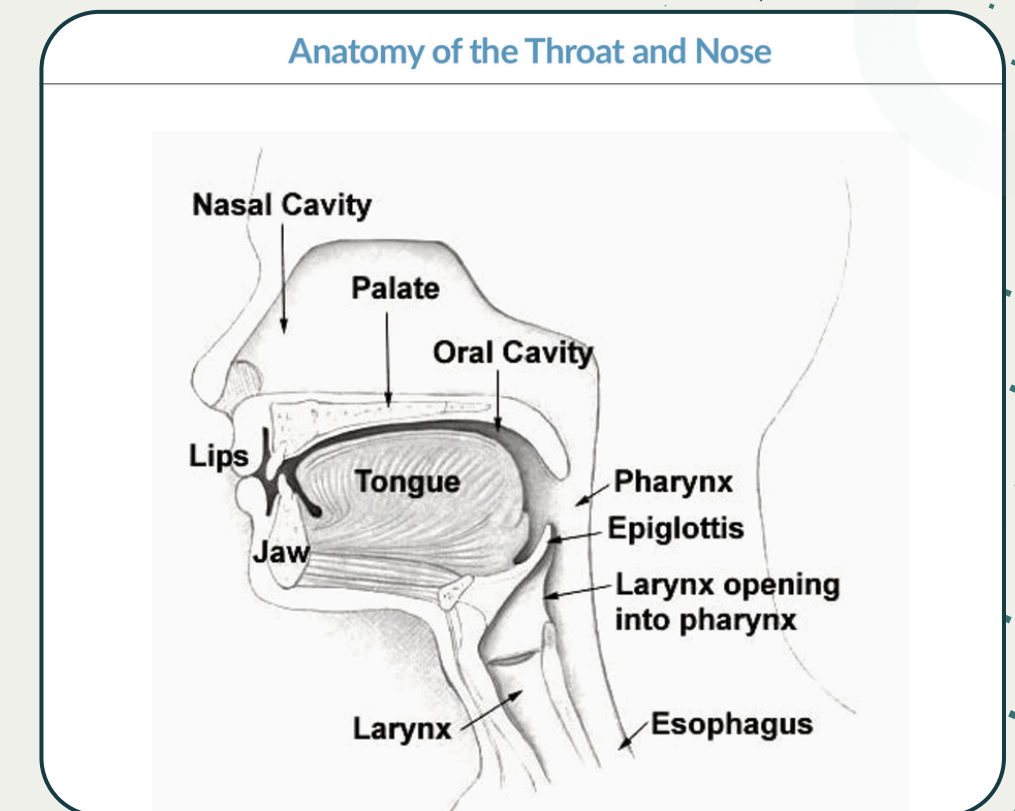
Baby Shark Fund Pitch

Dysphagia

Difficulty swallowing

Patients often need texture-modified meals.

Safe, but often look unappetising, lower intake and loss of dignity.



Why this matters?

>50% of older nursing home residents have dysphagia

>75% of SGH inpatients referred for swallowing assessment are
over 60

~50% of Singapore's population aged 65+ by 2050

Problem Statement

How might we use 3D food printing to make dysphagia meals visually appealing enough that elderly patients actually want to finish them?

Fix the problem of...

Unappealing

Food looks unappetising,
reducing patient appetite &
intake

Loss of dignity

Patients feel embarrassed eating
different food from others

Malnutrition risk

Incomplete meals lead to
nutritional deficiencies

What is MakanPrint?

3D Printed

Extrusion based, on demand

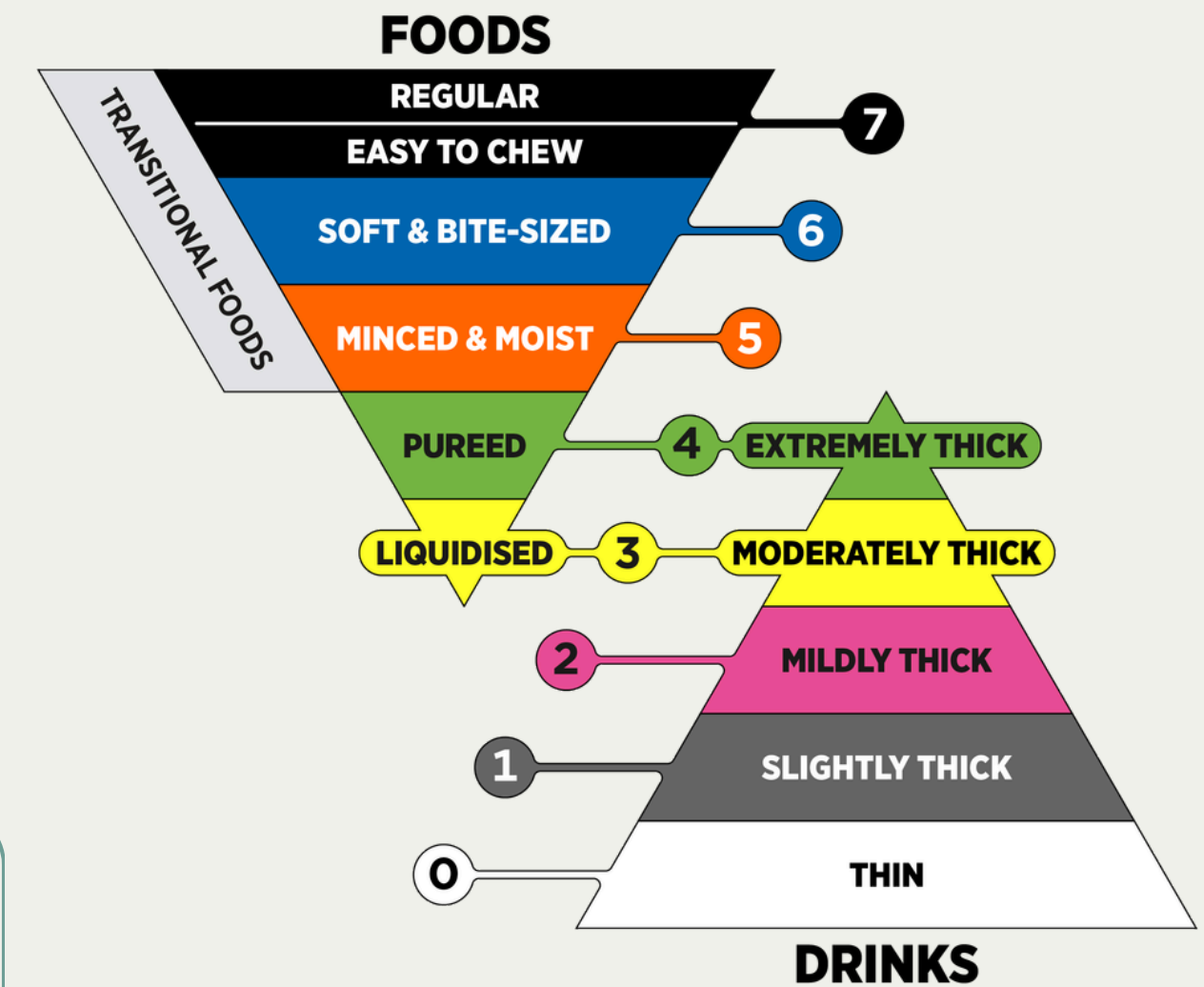


IDDSI Level 4–5

Clinically compliant

Asian & Halal

Cultural Relevance



Prior Prototype

Technical Feasibility from TIIDE



Formulation

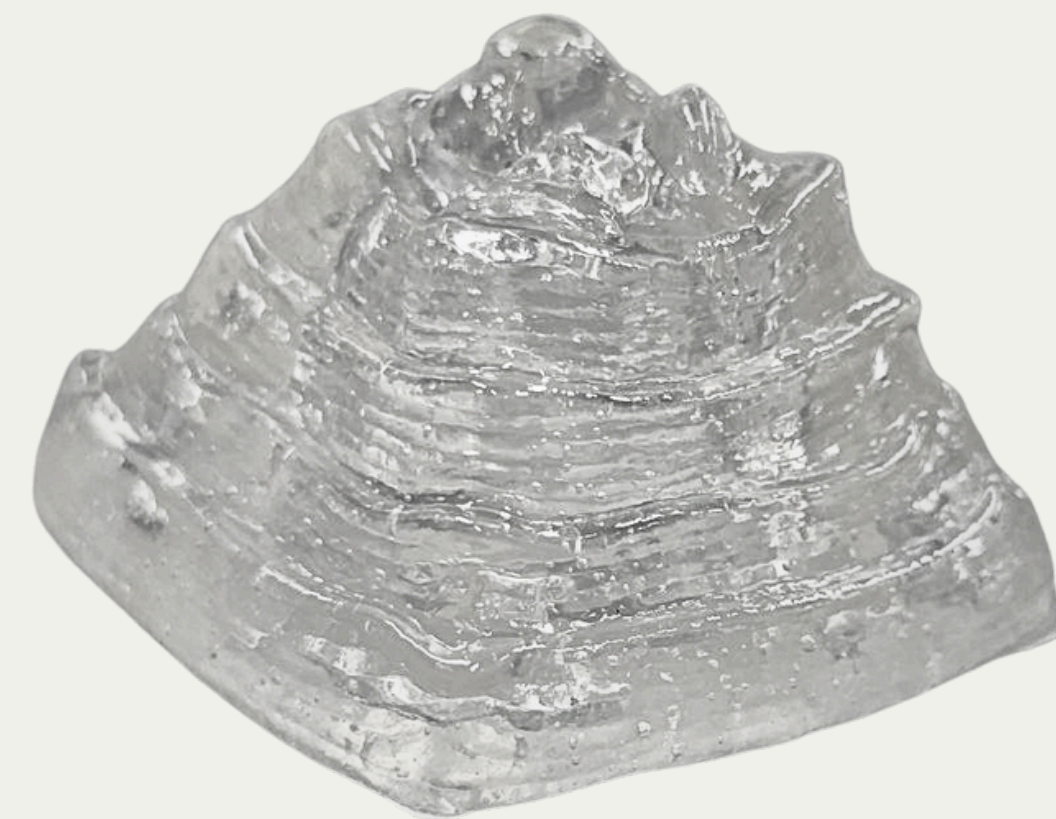


Printing

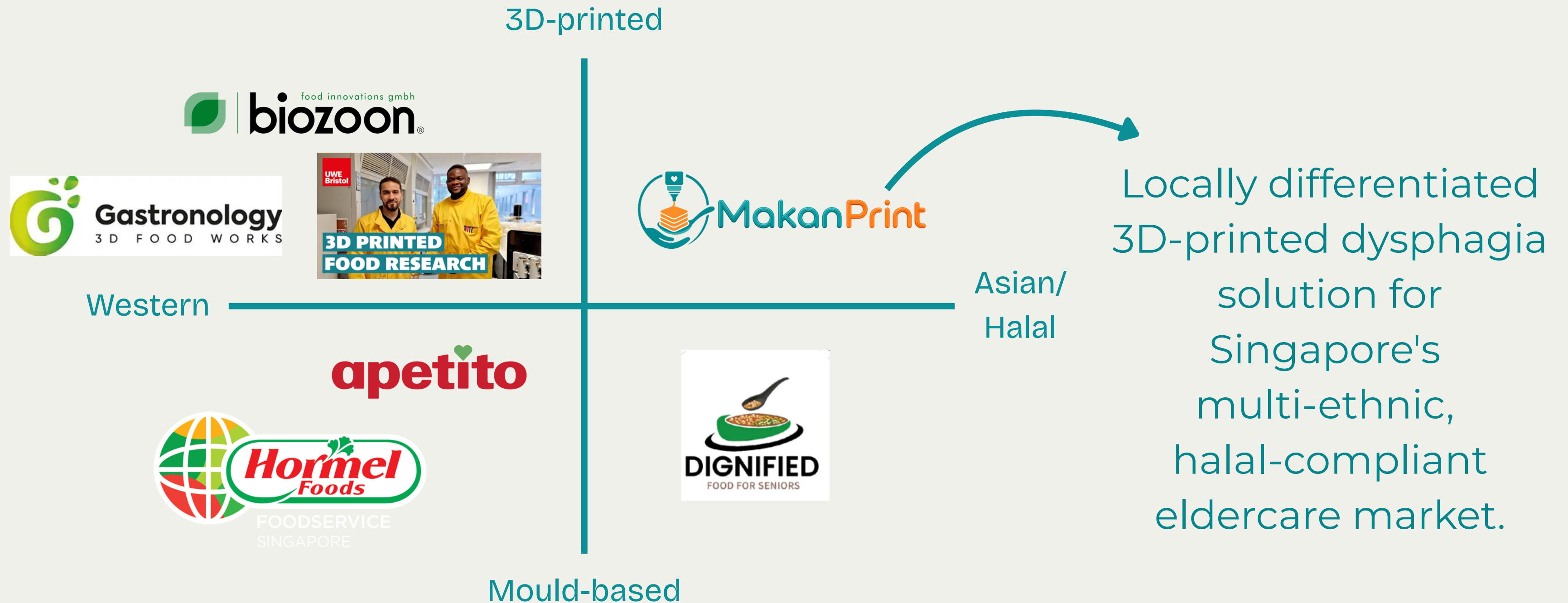
Validation



Prototype v0



How do we stand out



Design + AI

Bayesian optimisation for ink formulation

reduce experimental runs
required to get formulation

Computer vision for IDDSI compliance grading

automate fork pressure test
used in IDDSI grading

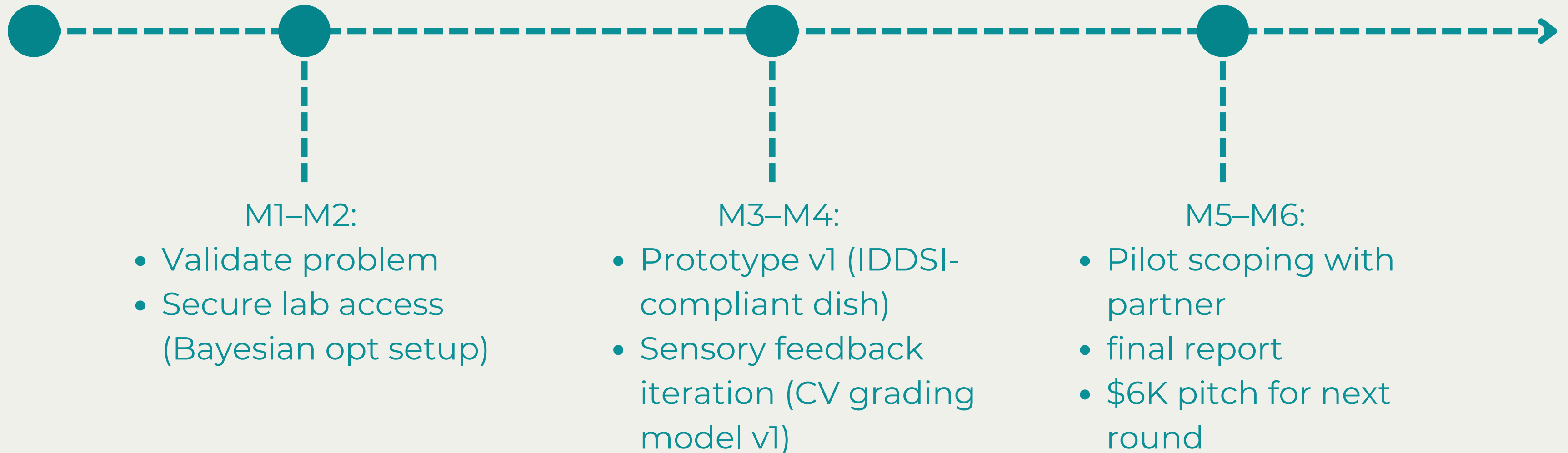
AI accelerates formulation testing and supports IDDSI screening.
Final validation remains clinician-led.

Outreach pipeline

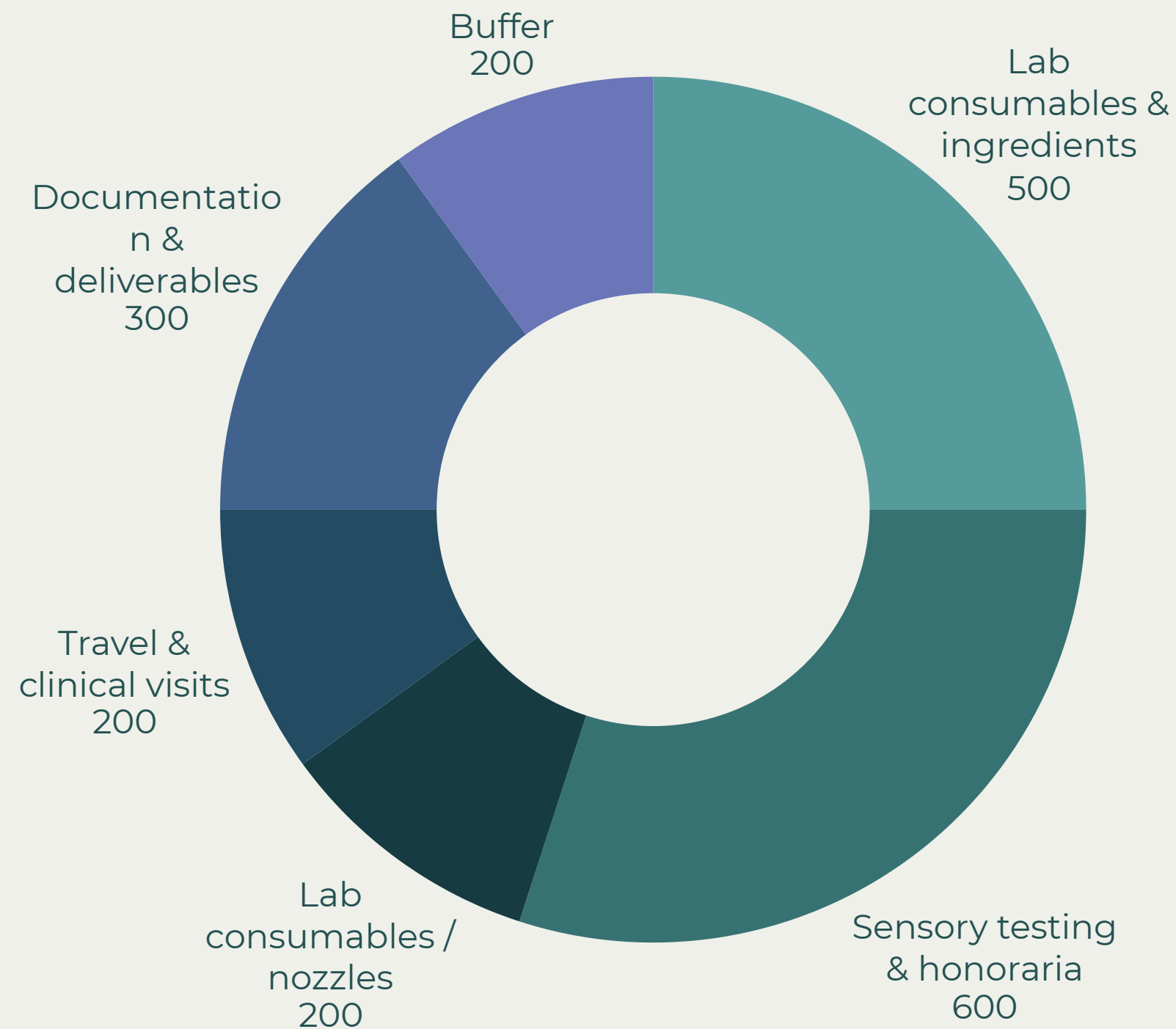
Stakeholders	Status
Changi General Hospital (Primary clinical pilot conversation)	Conversation initiated (via team gastroenterologist contact)
SUTD 3D Food Printing Lab	Outreach scheduled
Eldercare pilot site (operational testing)	To be confirmed

Our 6-month plan

Now



Breakdown of 2k fund



\$2,000 covers the early-stage validation and prototyping phase. Outcome at 6 months: 2–3 IDDSI-compliant ink formulations, sensory validation data, and the partnership and prototype evidence needed to pitch the \$6K track

Our Team

Zhenhao

ESD, Optimisation
lead + Outreach

Chao Ran

CSD, Computer vision
Lead

Lavanya

CSD, dataset and
training pipeline +
Outreach

Caitlyn

EPD, 3DP Lead and
lab work

Hillary

ESD, Optimisation and
Analytics Lead

Ryan

ESD, Optimisation
and project ops



**Every meal is a
chance
to restore dignity**

Baby Shark Fund Pitch 2026

Appendix

Citations

United Nations, Department of Economic and Social Affairs, Population Division. (2017). World Population Ageing 2017 — Highlights (ST/ESA/SER.A/397). Source for: "50% of Singapore's population projected to be 65+ by 2050"

Wong SM. (2019). Managing Dysphagia in an Ageing Society. activeage.co. Citing Singapore General Hospital internal data, 2008–2018. Source for: ">75% of SGH inpatients referred for swallowing assessment are over 60"

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Cichero JAY et al. (2017). Development of International Terminology and Definitions for Texture-Modified Foods and Thickened Fluids Used in Dysphagia Management: The IDDSI Framework. *Dysphagia*, 32, 293–314.

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Gastronology / Dysphalicious commercial launch — 3DPrint.com, April 2024.

Apetito Texture Modified range — apetito.co.uk.

Hormel Health Labs / Thick and Easy — hormelhealthlabs.com.

Health Food Matters × Changi General Hospital partnership — National Health Innovation Centre case study (2016).

DIGNIFIED / MAKANsoft, Singapore Institute of Technology — SilverStreak and Bakery & Snacks coverage (May 2026).

Snoek J, Larochelle H, Adams RP. (2012). Practical Bayesian Optimization of Machine Learning Algorithms. NeurIPS. (Foundational reference for Bayesian optimisation methodology.)

Pant A et al. (2022) — also relevant here as the prior SUTD work using classical RSM that our Bayesian approach extends.

Cichero et al. (2017) IDDSI framework — for fork-pressure test methodology that the CV model automates.